

Lab 7

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EE 446

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Submission

See submission files in ./submission folder.

Part I: Edge Impulse

Public project: <https://studio.edgeimpulse.com/public/1091500/latest> (*Motion Classification for Transport*, id 1091500).

Data collection

20 `idle` samples (board flat and still) and 20 `lift` samples (board in hand, repeated lift/lower), 10 s each, inertial sensor at 62.5 Hz over 6 axes. Rebalanced into 33 training / 7 test samples.

Impulse

The screenshot displays the Edge Impulse web interface for a project titled "Motion Classification for Transport" by user "vkutsyn". The interface is divided into a left sidebar and a main workspace. The sidebar includes navigation links for Dashboard, Data acquisition, Experiments, EON Tuner, Impulse design (with a sub-menu for Create impulse, Spectral features, Classifier, Retrain model, Live classification, Model testing, and Deployment), and a "GETTING STARTED" section with links to Documentation, Forums, and Discord. The main workspace is titled "Impulse #1" and contains a description: "An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data." Below this, four configuration panels are visible: 1. "Time series data" (red background) with settings for Input axes (6), Window size (2,000 ms), Window increase (stride) (500 ms), Frequency (Hz) (62.5), Zero-pad data (checked), and Train on data subset (100%). 2. "Spectral Analysis" (white background) with Name "Spectral features" and Input axes (6) including accX, accY, accZ, gyrX, gyrY, and gyrZ. 3. "Classification" (blue background) with Name "Classifier", Input features including Spectral features (checked), and Output features "2 (idle, lift)". 4. "Output features" (green background) showing "2 (idle, lift)". At the bottom of the workspace, there is a copyright notice "© 2026 Edgeimpulse Inc. All rights reserved." and a help icon.

Figure 1: Impulse: 2000 ms window @ 62.5 Hz → spectral analysis → classification.

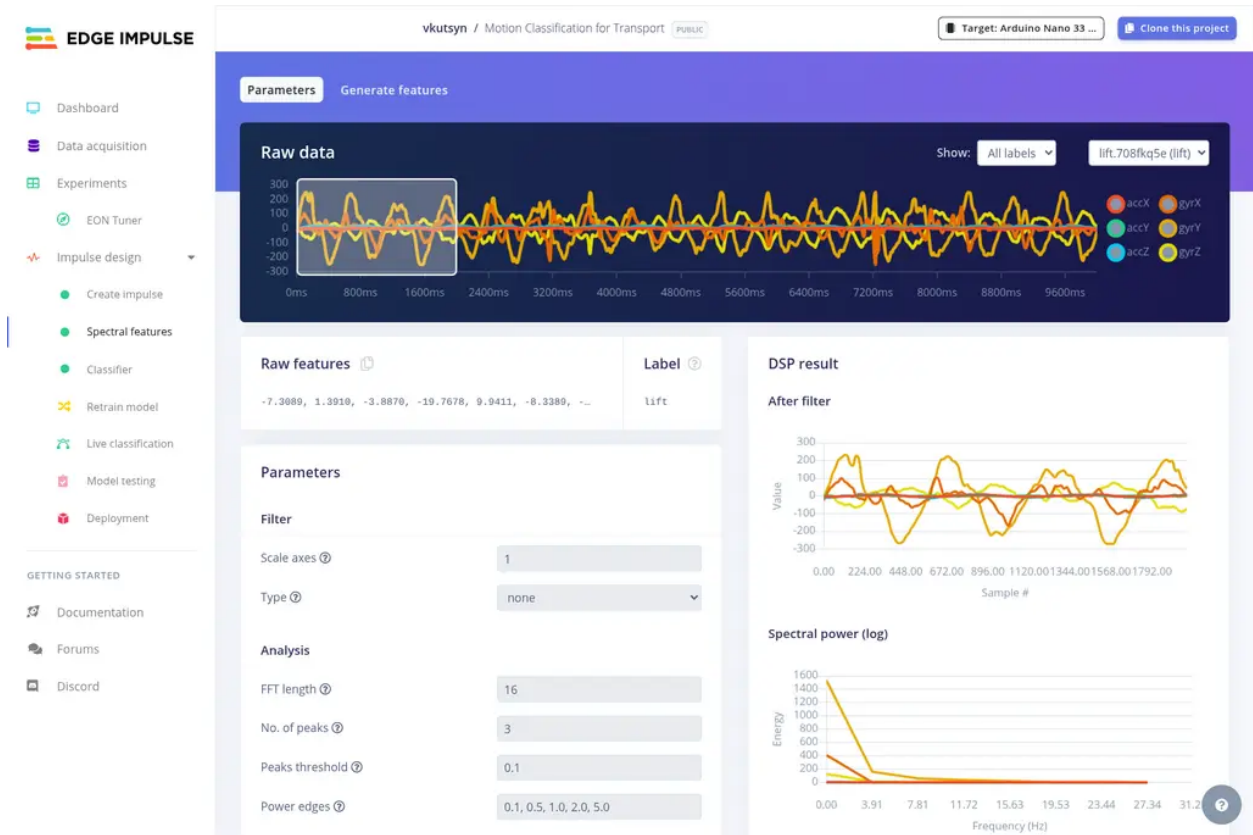


Figure 2: Spectral features for the six IMU axes.

Time-series input, 2000 ms window, 500 ms increase (561 windows), spectral analysis on all six axes (66 features), Keras classifier with 2 classes.

Training

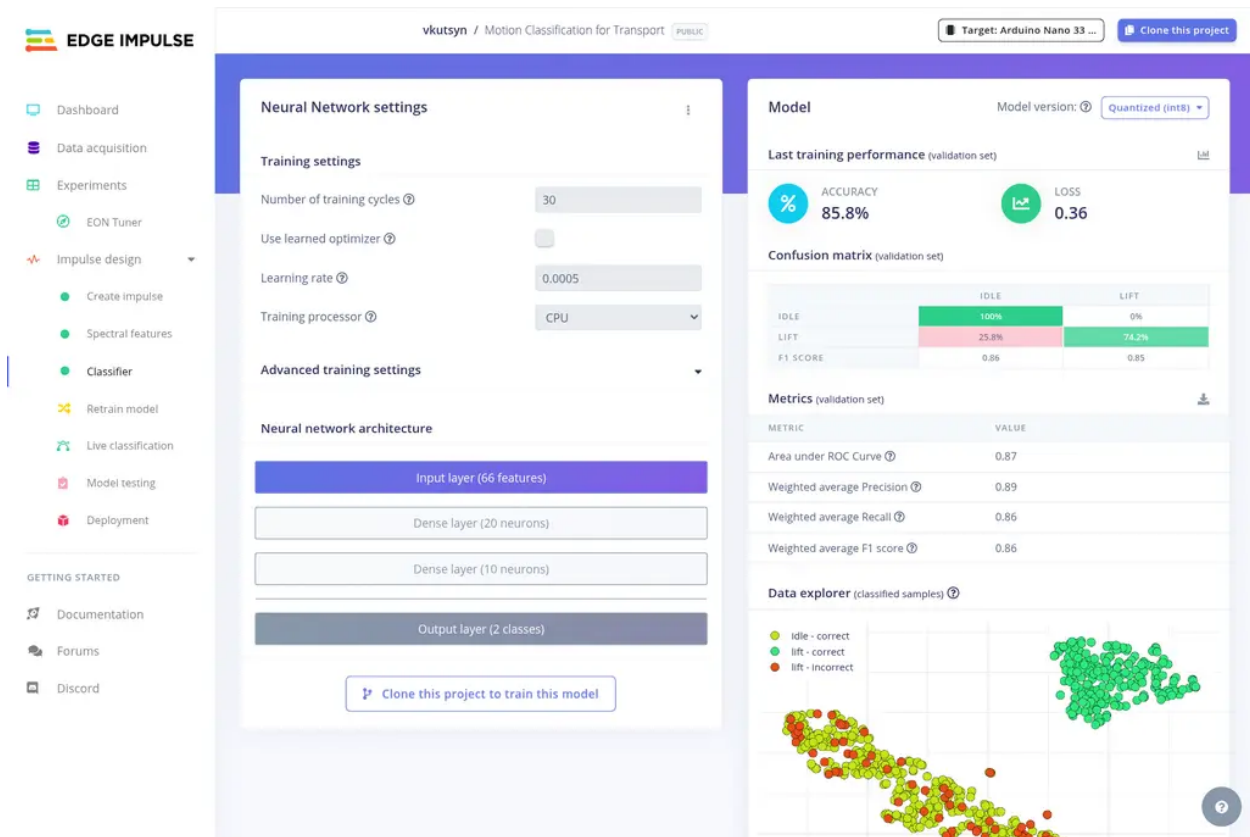


Figure 3: Network settings and validation results: 85.8 %.

66 → Dense(20) → Dense(10) → Dense(2), 30 cycles, lr 0.0005, 20 % validation split, int8. Validation accuracy 85.8 % for both float32 and int8, loss 0.36.

Test set

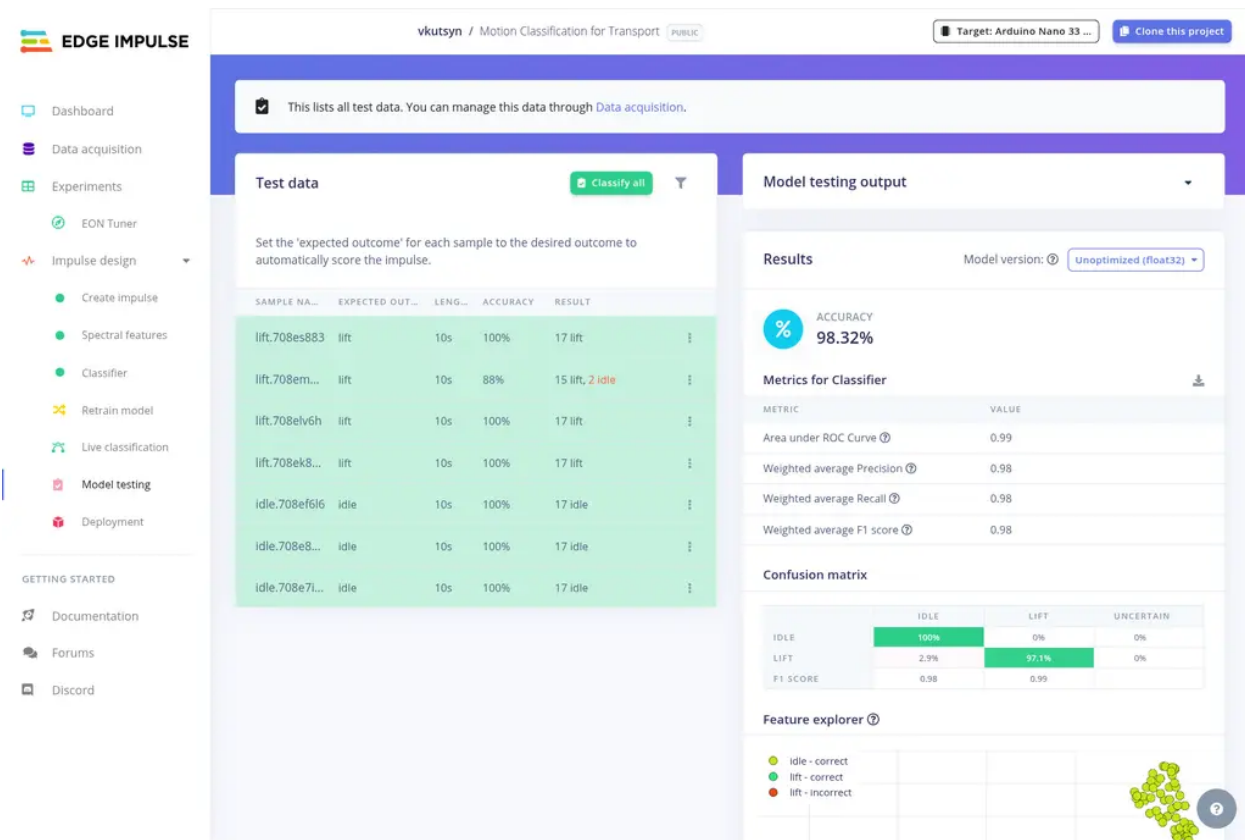


Figure 4: Model testing: 98.32 %.

98.32 % on the 7 held-out samples: idle 100 %, lift 97.1 %, AUC 0.99, F1 0.98.

Build

EON Compiler, int8, Nano 33 BLE Sense: 304 048 B flash (30 %), 58 848 B SRAM (22 %), DSP 33 ms + inference 0.63 ms per window. Arduino library and flashable firmware are in ./submission/part1.

On-device test

```
Timing: DSP 33 ms, inference 619 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 33 ms, inference 609 us, anomaly 0 ms, postprocessing 48 us
#Classification predictions:
  idle: 0.277344
  lift: 0.722656
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 33 ms, inference 611 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.753906
  lift: 0.246094
Starting inferencing in 2 seconds...
```

Figure 5: Serial output of `edge-impulse-run-impulse` on the flashed board.

`lift` wins while the board is being moved (0.996, then 0.723 as the motion stops) and `idle` wins once it is flat and still (0.754). Both states are identified correctly, with one window (2 s) of lag on the switch. Full log in `submission/part1/serial_inference_log.txt`.

Part II: Tiny Ensemble Learning

Question 1: Model architecture and ensemble flow

Each sample is a 100-time-step window of 6 IMU channels flattened row-major into $100 \times 6 = 600$ input neurons. The same window is fed to three branches that differ only in the input scaling applied to the raw signal: raw (unscaled, m/s² and deg/s), standard-scaled (`StandardScaler`, per-channel zero mean / unit variance) and min-max-scaled (`MinMaxScaler`, per-channel to $[0, 1]$).

Per branch (identical topology, three independently trained copies):

- Autoencoder used only for representation learning, 50 epochs, Adam(1e-3), MSE: encoder $600 \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(32, \text{linear})$, decoder $32 \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(600, \text{linear})$. The decoder is discarded after training; only the encoder is kept.
- Classifier on the frozen 32-D latent vector, 50 epochs, Adam(1e-3), categorical cross-entropy: $32 \rightarrow \text{Dense}(20, \text{ReLU}) \rightarrow \text{Dense}(12, \text{Softmax})$.

For deployment each branch is flattened into one Sequential model: $600 \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(32, \text{linear}) \rightarrow \text{Dense}(20, \text{ReLU}) \rightarrow \text{Dense}(12, \text{Softmax})$.

The ensemble contains 7 trained networks: 3 autoencoders (encoders kept), 3 branch classifiers and 1 meta-classifier; at inference time 4 models run (3 flattened branches + meta). Each branch emits a 12-element softmax vector; the three are concatenated into $3 \times 12 = 36$ values in the fixed order `[raw | std | minmax]`. The stacked meta-classifier is $36 \rightarrow \text{Dense}(24, \text{ReLU}) \rightarrow \text{Dense}(12, \text{Softmax})$, and its argmax is the final activity.

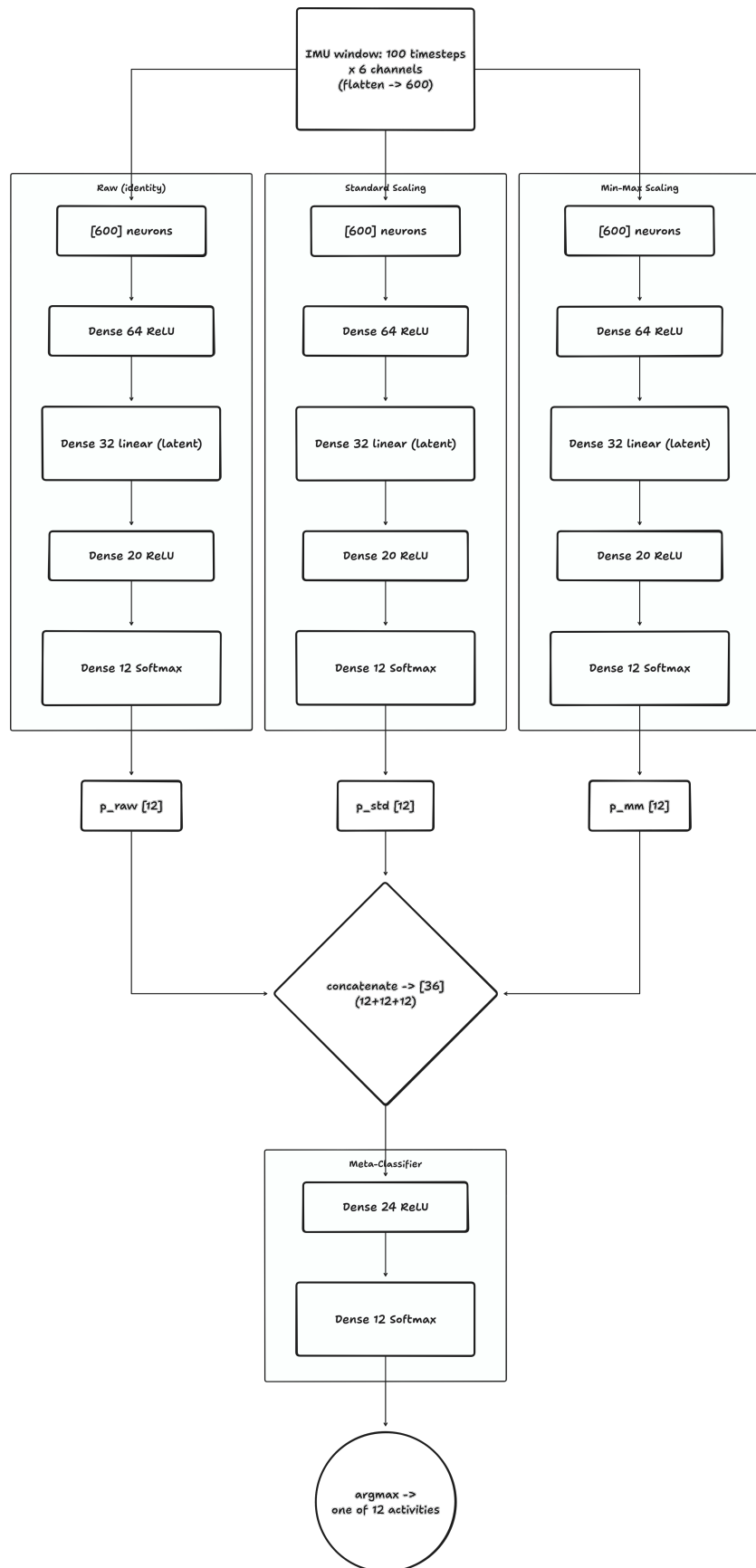


Figure 6: Diagram.

The 12 output classes are the mHealth activities: standing still, sitting and relaxing, lying down, walking, climbing stairs, waist bends forward, frontal elevation of arms, knees bending, cycling, jogging, running, jump front and back (labels shifted from 1-based to 0-based).

Question 2: Pruning and QAT methodology

Pruning

Each flattened branch and the meta-classifier are wrapped with `prune_low_magnitude` and fine-tuned for 5 epochs with the `UpdatePruningStep` callback. The schedule is `PolynomialDecay(initial_sparsity=0.10, final_sparsity=0.80, begin_step=0, end_step=500)`: sparsity ramps smoothly from 10 % to 80 % over the first 500 training steps, so the smallest-magnitude weights are zeroed gradually while the remaining weights keep adapting, instead of one destructive one-shot cut.

Stripping before QAT

`strip_pruning` removes the `PruneLowMagnitude` wrappers, the mask variables and the pruning step counters, leaving a plain Keras model whose kernels simply contain zeros. This is required because `quantize_model` cannot wrap an already-wrapped layer — nesting quantize wrappers around pruning wrappers is unsupported, and the leftover pruning bookkeeping variables would also end up in the converted graph. Stripping keeps the sparsity (the zeros) but drops the machinery.

QAT after pruning

The stripped sparse model is passed through `quantize_model` inside a `quantize_scope()`, which inserts fake-quant nodes on weights and activations, then it is fine-tuned for 5 more epochs. The network therefore learns weights that are already robust to int8 rounding, recovering most of the accuracy that naive post-training quantization would lose.

Why the extra mask enforcement

After stripping, nothing constrains the zeros any more: QAT is ordinary gradient descent, so Adam updates (momentum, weight decay, non-zero gradients) immediately push pruned weights off zero, and the fake-quant rounding can also map near-zero values to a non-zero quantization level. The sparsity would silently evaporate during the 5 QAT epochs. The notebook therefore snapshots the binary masks (`extract_dense_weight_masks, w != 0`) from the stripped model and re-applies them with `MaskEnforcerCallback` after every training batch and every epoch, plus once more after `fit` returns, so the exported model provably keeps ~80 % zeros (`compute_model_sparsity` prints the layer-wise result).

int8 TFLite conversion

The QAT model is converted with `optimizations = [Optimize.DEFAULT]`, `supported_ops = [TFLITE_BUILTINS_INT8]` and `inference_input_type = inference_output_type = tf.int8`, i.e. a full-integer model with no float fallback. A `representative_dataset` of 200 evenly spaced training windows is required because weight ranges are known statically but **activation** ranges are not: the converter runs those samples through the graph to observe each intermediate tensor's dynamic range and derive its scale and zero-point. Without representative data the converter

cannot calibrate activations and would fall back to a dynamic-range or hybrid model, which the TFLITE_BUILTINS_INT8 target forbids.

Why this matters on the Nano 33 BLE Sense

The board has 256 KB SRAM and 1 MB flash with no FPU-friendly float kernels. int8 quantization gives a 4x reduction in weight storage and lets the CMSIS-NN integer kernels run the MACs, which is both faster and lower-energy than float emulation. Measured here, each branch drops from ~169 KB of float32 TFLite (163.7 KB encoder + 5.4 KB classifier) to **45.6 KB** int8, and the meta-classifier from 6.5 KB to **3.6 KB** — all four models together are ~139 KB, which fits in flash alongside a 56 KB tensor arena. Pruning contributes robustness and compressibility (the 80 % zeros compress well in the .cc array and shrink the effective model), while quantization delivers the direct memory and latency win.

Question 3: Arduino deployment and real-world behavior

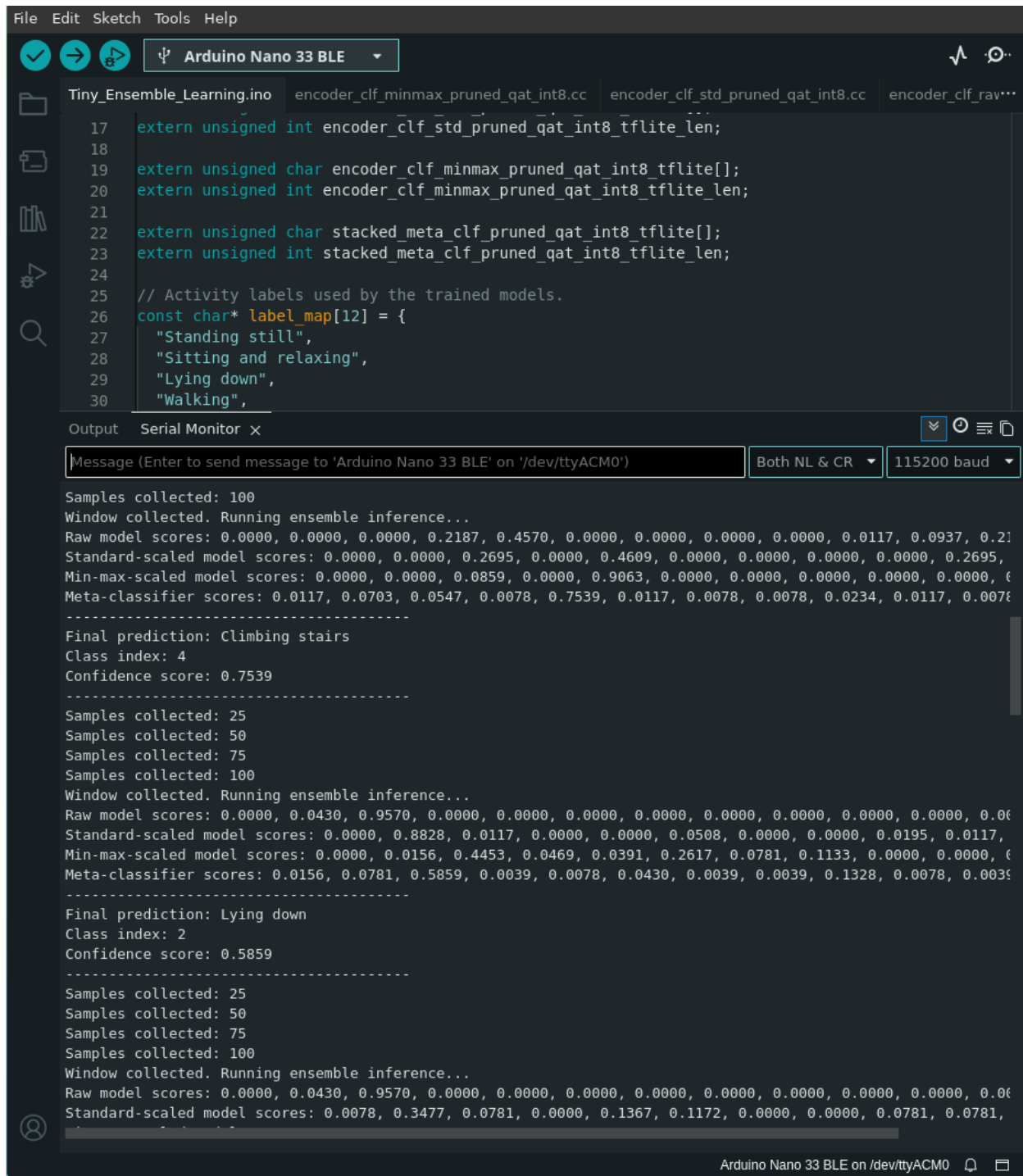


Figure 7: Serial monitor output with the board resting flat

With the board lying flat and motionless, the sketch collects 100 samples at 50 Hz (2 s per window) and then prints the three branch softmax vectors and the stacked prediction.

Issues:

Unit mismatch on the gyroscope

mHealth stores gyro in deg/s, but the sketch multiplies `readGyroscope()` (deg/s) by `kDegToRad`, so on-board gyro values are ~57x smaller than anything the models saw in training.

Sensor placement / orientation mismatch

The six columns used from the log are the *left-ankle* accelerometer and gyroscope. The training means (`mean(Accel_Y) = -9.15 m/s2`) show gravity on the Y axis, whereas a Nano lying flat on a table puts -9.8 m/s^2 on Z. The static gravity signature is on the wrong axis before any motion is even considered.

Practical improvements:

1. Fix the units and axes first — drop the deg/s to rad/s conversion, and remap or re-orient the board axes so gravity lands on the same channel as in training (or mount the board on the ankle as in the dataset).
2. Collect a small calibration set on the board itself with `Raw_IMU_Recorder`, then either recompute the scaler statistics from that data or fine-tune the models on it; additionally smooth the output with a majority vote over several consecutive windows and reject predictions below a confidence threshold.

Question 4: Deployment behavior on the Arduino

Yes — the deployed ensemble repeatedly predicts the same class even when the board is moved differently, and the meta softmax stays near 1.0 for that class.

The root cause is domain shift: the models were trained on the mHealth subject-6 left-ankle Shimmer channels and are being fed BMI270 data from a hand-held board.

The fix is to align the deployment domain with the training domain (correct units and axes, matching sensor placement) or, better, to retrain on data recorded from the board itself.